

GRANT COUNTY REGIONAL, OR, USA

WMO: 726876

Lat: 44.400N

Lon: 118.967W

Elev: 3698

StdP: 12.84

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00387

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 10.1	(c) 16.0	(d) 3.3	(e) 7.4	(f) 16.3	(g) 8.8	(h) 9.9	(i) 24.3	(j) 32.1	(k) 45.0	(l) 27.6	(m) 43.2	(n) 5.8	(o) 270	(p) 0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.7	94.8	60.7	91.4	59.9	88.8	59.2	63.9	87.3	62.1	85.4	60.7	83.7	8.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.0	76.6	66.5	54.1	71.5	65.7	52.1	66.4	65.3	31.0	87.3	29.6	84.8	28.6	84.2	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.0	(b) 19.1	(c) 16.4	(e) 2.9	(f) 99.9	(g) 5.2	(h) 3.4	(i) -0.8	(j) 102.3	(k) -3.8	(l) 104.3	(m) -6.7	(n) 106.3	(o) -10.5	(p) 108.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			2.9	99.9	5.2	3.4	-0.8	102.3	-3.8	104.3	-6.7	106.3	-10.5	108.8		
			1.8	66.6	6.1	2.2	-2.5	68.1	-6.1	69.4	-9.5	70.6	-13.9	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	35.0	37.3	41.8	46.5	54.3	61.9	72.0	70.8	62.8	50.9	41.0	32.8
	DBStd	15.19	8.01	7.72	6.77	7.07	7.68	7.67	5.54	6.05	8.02	7.68	8.81	8.71
	HDD50	2201	465	358	264	152	44	6	0	0	6	82	289	535
	HDD65	5793	929	776	719	557	339	145	11	21	134	442	721	999
	CDD50	2444	1	2	10	46	178	363	681	643	391	109	19	1
	CDD65	562	0	0	0	0	9	52	228	200	69	4	0	0
	CDH74	8385	0	0	1	33	225	896	3251	2816	1091	72	0	0
	CDH80	3785	0	0	0	2	51	325	1609	1346	432	20	0	0
Wind	WSAvg	6.5	7.1	7.1	7.5	6.9	6.0	5.9	5.8	5.7	5.8	6.3	7.1	7.2
Precipitation	PrecAvg	14.8	1.6	1.1	1.3	1.6	1.7	1.4	0.6	0.4	0.6	1.2	1.7	1.7
	PrecMax	19.7	3.0	2.3	3.1	2.6	3.9	2.8	1.7	2.0	1.7	2.4	3.0	4.1
	PrecMin	9.0	0.5	0.4	0.6	0.9	0.0	0.2	0.0	0.0	0.0	0.4	0.5	0.6
	PrecStd	2.8	0.7	0.4	0.7	0.5	1.0	0.7	0.5	0.5	0.4	0.5	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.2	67.7	79.0	85.7	94.4	98.2	99.0	93.4	82.4	68.1	55.3
		MCWB	43.3	45.4	49.0	54.2	60.8	63.8	62.1	61.8	59.5	55.6	51.3	45.7
	2%	DB	51.8	55.9	61.9	72.4	80.7	88.1	95.1	94.6	88.8	73.4	62.2	51.8
		MCWB	42.1	44.4	46.0	52.2	57.1	61.1	60.7	60.2	58.1	52.3	48.6	43.6
	5%	DB	48.1	52.2	57.5	66.2	75.1	82.9	92.6	90.9	85.7	69.3	57.7	47.7
		MCWB	41.2	43.1	44.7	49.4	55.2	58.9	60.3	59.3	57.1	50.7	46.8	41.8
	10%	DB	45.6	48.4	54.1	60.6	70.6	79.1	89.8	88.0	80.8	64.0	53.7	44.8
		MCWB	39.5	40.9	43.8	46.8	53.2	56.8	59.9	58.6	55.5	49.2	44.9	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	47.8	51.1	56.0	62.0	66.2	66.4	64.6	62.5	57.6	52.9	46.9
		MCDB	52.8	56.5	62.7	74.6	82.6	88.8	91.5	90.4	82.9	74.3	64.3	53.5
	2%	WB	43.7	45.8	48.4	52.8	58.9	63.0	64.2	62.5	60.4	54.9	50.5	44.5
		MCDB	49.0	54.3	57.9	69.8	74.9	83.6	88.7	86.9	83.0	68.0	59.5	50.1
	5%	WB	41.7	43.7	46.4	50.3	56.7	60.4	62.3	61.1	58.6	52.6	48.2	42.2
		MCDB	47.1	50.5	55.7	63.9	71.4	79.6	86.7	85.3	80.2	65.0	56.6	47.0
	10%	WB	40.0	41.6	44.4	47.9	54.7	58.3	60.8	59.7	56.7	50.8	45.6	39.7
		MCDB	45.0	48.0	53.0	58.4	67.7	75.7	84.9	83.6	77.6	61.6	52.6	44.0
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.6	18.1	20.8	23.5	26.0	31.7	31.2	28.2	21.1	15.5	12.2
		MCDBR	17.2	19.9	24.7	31.3	31.2	32.9	35.0	35.2	34.4	29.5	21.6	14.1
		MCWBR	10.5	11.4	13.0	15.2	13.8	13.0	12.5	12.6	13.6	13.5	11.5	8.8
	5% WB	MCDBR	14.9	17.7	21.0	28.2	27.3	30.4	31.8	32.0	31.5	24.5	19.1	12.9
		MCWBR	10.1	10.8	11.9	14.4	12.5	12.3	12.5	11.8	12.7	12.4	10.8	8.7
Clear-Sky Solar Irradiance	taub		0.282	0.286	0.298	0.350	0.355	0.340	0.346	0.344	0.325	0.301	0.283	0.279
	taud		2.487	2.531	2.527	2.353	2.375	2.449	2.451	2.449	2.487	2.563	2.557	2.511
	Ebn at Noon		270	291	300	289	290	294	291	288	286	281	269	260
	Edh at Noon		22	25	29	37	37	35	35	33	30	24	20	19
All-Sky Solar Radiation	RadAvg		498	783	1153	1626	1936	2274	2446	2093	1586	1006	567	390
	RadStd		63	88	119	136	164	126	97	122	102	85	50	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	N/A

Nomenclature: See separate page